

In-Class Lecture 11. Survival Analysis

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Overview

- 1 Life Tables and Estimation of the Survivor Function
- 2 Kaplan-Meier Estimator

Notation

The basis of survival analysis is the survival function:

$$S(t) = \Pr(T > t)$$

In this function, T is the time of the “event.” Also called “failure” or “death”.

And t is a specified time. So this is the probability that the time of death/failure (T) is later than the specified time (t).

For example, $S(2)$ is the $\Pr(T > 2)$, or the probability that death occurs after time 2.

Example 1

We have a sample size of 10. We have the following observed data:

Table: Example 1 Data

Case ID	Time of Event (T)
1	5
2	6
3	8
4	18
5	16
6	14
7	7
8	2
9	4
10	9

Further Notation

The distribution of T has the following **probability density function**:

$$f_T(t) \geq 0 \text{ for } t \geq 0$$

In this form, we are talking about the probability of failure or death (that's the event which occurs at time T).

We assume $T \geq 0$. We also start from the assumption that T is continuous.

Further Notation

We often look at the **cumulative** distribution function:

$$F_T(t) = P(T \leq t) = \int_0^t f_T(u) du$$

Again, this is the cumulative probability that death occurs before time t .

Survival Function

Recall the survival function:

$$S_T(t) = \Pr(T > t)$$

This is the probability that a person lives beyond time t .

This is the complement of the CDF we just looked at:

$$S_T(t) = \Pr(T > t) = 1 - \Pr(T \leq t) = 1 - F_T(t)$$

Further Notation

We often assume a parametric form for the CDF. This is denoted $F_T(t|\theta)$.

For example, we might assume that $T \sim N(\mu, \sigma^2)$. In this case, $\theta = (\mu, \sigma^2)$.

Nonparametric

We will focus on **nonparametric** estimates.

Parametric models give us convenient summaries with few parameters.

But does the model with few parameters fit the data well?

Hazard Function

We can also look at conditional survival probabilities.

These are summarized by the hazard function.

The proportion of those alive at time t that will die before time $t + 1$.

Since time is continuous, we can add any amount of time: Δt .

Hazard Function

In mathematical notation: $P\{t \leq T < t + \Delta t | T \geq t\}$

Since time is continuous, we speak of a limiting hazard rate:

$$h_T(t) = \lim_{\Delta t \rightarrow 0} \frac{P\{t \leq T < t + \Delta t | T \geq t\}}{\Delta t}$$

It can be shown that the former is equal to:

$$h_T(t) = \frac{f_T(t)}{S_T(t)} = \frac{f_T(t)}{1 - F_T(t)}$$

Summary of Functions

Each of these functions completely describe the distribution of T .

Any of these can be transformed into the other forms (see the notes).

Life Tables

Table: Typical Data for Life Table Analysis

Time Interval (Months)	At Risk	Deaths
[0, 2)	300	195
[2, 4)	105 (=300-195)	27
[4, 6)	78	15
[6, 8)	63	12
[8, 10)	51	9
[10, 12)	42	12
[12, 14)	30	

Life Tables

In this example, data are broken up into discrete intervals (even though time is continuous).

This is quite common. Many studies only observe units at specified points in time (e.g. every 2 months).

This leads to some issues. If we observe death has occurred by $t = 6$, we only know if occurred after $t = 4$ and some time before $t = 6$.

In some problems, the “time” dimension is actually discrete (e.g. calls).

Life Table Notation

Notation	Description
x	time at beginning of interval
O_x	number <i>at risk</i> at exact time x
n	length of interval
${}_n d_x$	number dying in interval x to $x + n$
${}_n p_x$	probability of surviving from time x to time $x + n$
${}_n q_x$	probability of dying during time x to time $x + n$
${}_n P_x$	probability of surviving during time x to time $x + n$ (more than one interval)

Life Table Notation

Define the probabilities:

$${}_n\hat{q}_x = \frac{{}_nd_x}{O_x}$$

$${}_n\hat{p}_x = 1 - \frac{{}_nd_x}{O_x} = 1 - {}_n\hat{q}_x$$

Therefore ${}_n\hat{p}_x + {}_n\hat{q}_x = 1$. But ${}_n\hat{p}_x + {}_{n+1}\hat{q}_x \neq 1$.

Life Table Notation

The following is the notation for the probability that a unit will survive one year:

$$\begin{aligned} {}_{12}P_0 &= \prod_{i=0}^5 {}_2p_{2i} = ({}_2p_0) ({}_2p_2) ({}_2p_4) ({}_2p_6) ({}_2p_8) ({}_2p_{10}) \\ &= Pr(\text{Survive to 2} | \text{alive at 0}) \cdot Pr(\text{Survive to 4} | \text{alive at 2}) \cdot \dots \end{aligned}$$

In our example, ${}_4P_0 = \frac{300-195}{300} \cdot \frac{105-27}{105} = 0.35 \cdot 0.74 = 0.26$

In-Class Example

Table: In-Class Example

			Probability		
			Dying in interval	Surviving given alive at start of interval	Surviving to end of interval given alive at $t=0$
x	O_x	${}_1d_x$	${}_1\hat{q}_x$	${}_1\hat{p}_x$	${}_{x+1}\hat{P}_0$
0	100	10			
1	90	9			
2	81	12			

In-Class Example

Table: In-Class Example

			Probability		
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x	O_x	${}_1d_x$	${}_1\hat{q}_x$	${}_1\hat{p}_x$	${}_x+1\hat{P}_0$
0	100	10	0.1	0.9	0.9
1	90	9	0.1	0.9	0.81
2	81	12	0.148148	0.851852	0.69

Censoring

Define notation for withdrawals:

${}_nW_x$ number of withdrawals from time x to time $x + n$

This creates an issue. We don't know when the withdrawal occurred. We can assume they withdrew at the beginning, the end, or in the middle somewhere.

Censoring

It matters since it changes the size of the risk set, O_x .

If we assume the withdrawals occurred at the *end* of the interval, then the following estimator is close:

$${}_2\hat{q}_x \approx \frac{{}_2d_x}{O_x}$$

If we assume the withdrawals occurred at the *beginning* of the interval, then the following estimator is close:

$${}_2\hat{q}_x \approx \frac{{}_2d_x}{O_x - {}_2w_x}$$

Censoring

If we assume that withdrawals are uniformly distributed across the interval, then we might assume that the withdrawals faced half the risk as those who stayed the full interval.

Therefore, reduce the risk set by half the number of withdrawals.

$${}_2\hat{q}_x = \frac{{}_2d_x}{O_x - {}_2w_x/2}$$

In the notes, the denominator is re-written as $O'_x = O_x - {}_2w_x/2$.

Variance Estimator

$$\hat{V} \left\{ {}_{x+2}\hat{P}_0 \right\} = {}_{x+2}\hat{P}_0^2 \sum_{i=1}^{x/2} \frac{{}_2\hat{q}_x}{O_i - {}_2d_i}$$

Although this may not look like it, in the absence of withdrawals, this is a $\frac{pq}{n}$ kind of thing.

This variance estimator is known as Greenwood's formula.

Variance Estimator

This uses our two-month interval scheme. A more general formulation denotes each interval $i = 1$ to I . Observation starts at time 0, $i = 0$. Each time point where observations are taken is iterated, but the interval is arbitrary.

We can re-write our formulae using this more general notation for time:

$$\hat{p}_i = 1 - \frac{d_i}{O_i - w_i/2}$$

$$\hat{P}_I = \prod_{i=1}^I \hat{p}_i$$

$$\hat{V} \{ \hat{P}_I \} = \hat{P}_I^2 \sum_{i=1}^I \frac{\hat{q}_i}{O_i - d_i}$$

Kaplan-Meier Estimator

The nonparametric maximum likelihood estimator of the survivor function.

Exact when time of event and withdrawals are known, i.e. the risk set is known.

Notation

- i =specific survival time
- O_i =number at risk at time i
- d_i =number of deaths at time i
- ${}_n\hat{P}_0$ =survival function from time 0 through time n

Kaplan-Meier Estimator

Kaplan-Meier Estimator is:

$${}_n\hat{P}_0 = \prod_{i \leq n} \left(1 - \frac{d_i}{O_i}\right)$$

The variance estimator is:

$$\hat{V} \left\{ {}_n\hat{P}_0 \right\} = {}_n\hat{P}_0^2 \sum_{i=1}^n \frac{d_i}{O_i(O_i - d_i)}$$

Example 3

Two treatments for acute leukemia. The table lists times to event or censoring (the asterisk indicates cases that were censored).

Data for Example 3	
6-MP	6, 6, 6, 6*, 7, 9*, 10, 10*, 11*, 13, 16, 17*, 19*
	20*, 22, 23, 25*, 32*, 32*, 34*, 35*
Placebo	1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17
	22, 23

Example 3

Estimation of Survival Function for Example 3

Drug 6-MP				Placebo			
Time	O_i	d_i	${}_i\hat{P}_0$	Time	O_i	d_i	${}_i\hat{P}_0$
6	21	3	0.857	1	21	2	0.905
7	17	1	0.807	2	19	2	0.810
10	15	1	0.753	3	17	1	0.762
13	12	1	0.690	4	16	2	0.667
16	11	1	0.627	5	14	2	0.571
22	7	1	0.538	8	12	4	0.381
23	6	1	0.448	11	8	2	0.286
				12	6	2	0.190
				15	4	1	0.143
				17	3	1	0.095
				22	2	1	0.048
				23	1	1	0.000

In-Class Problem: Example 3

Fill in the $Var({}_i\hat{P}_0)$ column:

Table: Example 3 Data: Drug 6-MP

Time	O_i	d_i	${}_i\hat{P}_0$	$Var({}_i\hat{P}_0)$
6	21	3	0.857	
7	17	1	0.807	
10	15	1	0.753	

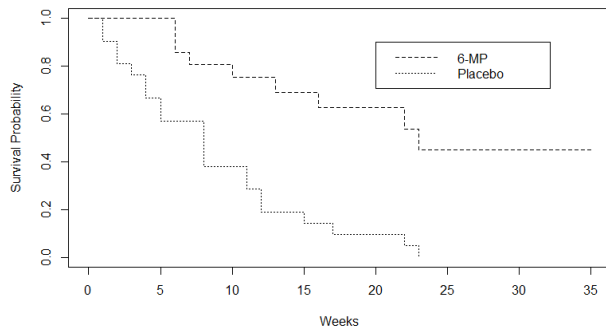
In-Class Problem: Example 3

Fill in the $Var({}_i\hat{P}_0)$ column:

Table: Add caption

Time	O_i	d_i	${}_iP_0$	$Var({}_i\hat{P}_0)$
6	21	3	0.857	0.005829
7	17	1	0.807	0.007563
10	15	1	0.753	0.009285

Example 3



Example 3

R code to get the KM estimator and Survival Plot:

```
#coding is for the event, 1=event, 0=censored
#The variable that I have is for censoring.
library(survival)
example3$event <-abs(example3$censored-1)

surv1 <- survfit(Surv(weeks, event)~ group,
  conf.type="none",data=example3)
surv1
summary(surv1)
plot(surv3, xlab="Weeks",
  ylab="Survival Probability",
  conf.int=FALSE,lty=2:3)
legend(20, .9, c("6-MP", "Placebo"),lty=2:3)
```